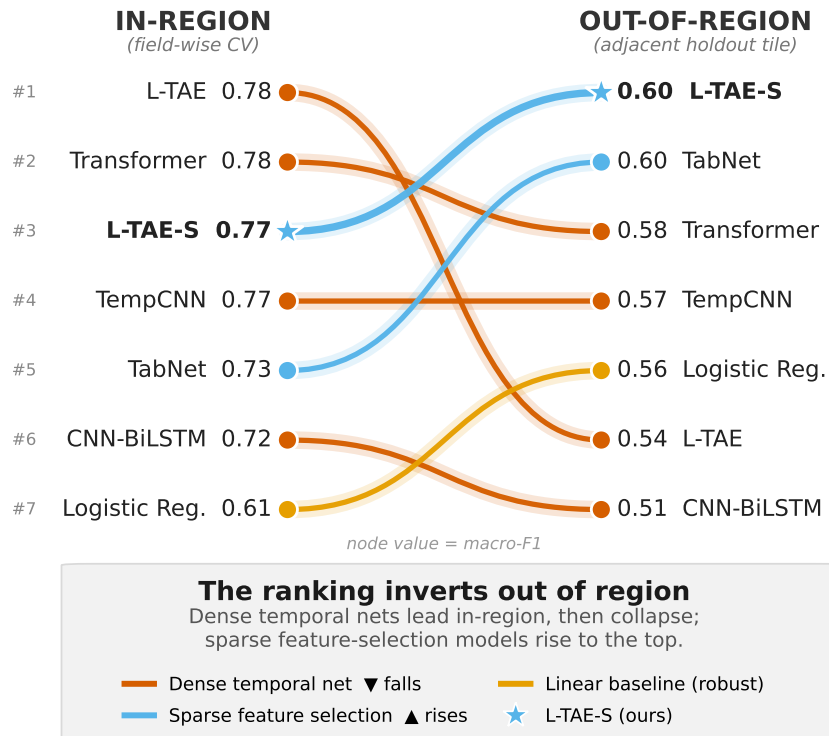


1 Graphical Abstract

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3 **of-Region Transfer in Crop Classification**

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Sparse Feature Selection Outweighs Deep Learning Capacity for Out-of-Region Transfer in Crop Classification

Michael L. Mann^{a,*}, Sairam Venkatachalam^b, Disha Kacha^b, Devarsh Sheth^b,
Ryan Engstrom^a, Amir Jafari^b

^a*Department of Geography & Environment, The George Washington
University, Washington, DC, 20052, USA*

^b*Data Science Program, The George Washington University, Washington, DC, 20052, USA*

Abstract

Crop-type maps from satellite image time series are a backbone of agricultural monitoring, yet the machine-learning models that produce them are almost always trained and tested within a single region, leaving open how they behave when deployed to new areas where no labels were collected. Using a South African Sentinel-2 dataset (optical only), we benchmark twenty-three classical machine-learning and deep-learning models under out-of-region transfer, applying them without any target-region labels or adaptation to an adjacent, non-overlapping holdout tile, and identify the model property most associated with success: sparse, axis-aligned feature selection. Models that commit to a few features at a time, gradient-boosted trees, logistic regression, and the attentive TabNet, transfer robustly (macro-F1 0.56–0.60), whereas high-capacity dense temporal networks (CNN-BiLSTM, a Transformer encoder, L-TAE) degrade sharply; in a controlled comparison, adding sparse feature selection to L-TAE improves transfer more than adding attention capacity does. We turn this insight into a method: L-TAE-S, a sparse-attention temporal encoder that ties the best model for out-of-region macro-F1 (0.60) at a fraction of the training cost, with a field-level variant that matches this accuracy while training in seconds per seed. We further show that conventional in-region validation inverts these rankings, the dense networks that lead in-region (macro-F1 0.72–0.78) fall furthest under transfer, which is why models intended for operational crop mapping should be evaluated outside the region they were trained on. Field-level aggregation is a second, substitutable robustness lever, and the most transferable models retain near-full accuracy using only a quarter of the training fields. Together these results give practitioners a concrete design principle for crop classifiers that must be applied to nearby unlabeled farmland.

Keywords: Crop classification, deep learning, feature selection, remote sensing, Sentinel-2, out-of-region transfer, time-series classification

*Corresponding author.

Email address: mmann1123@email.gwu.edu (Michael L. Mann)
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1. Introduction

Accurate and timely crop classification is foundational to precision agriculture and to the agencies that monitor food security and agricultural land use, enabling yield forecasting, resource optimization, and food-security assessment (Bégué et al., 2020; Ienco et al., 2017). The Sentinel-2 constellation, with its multispectral richness and roughly five-day revisit, has made dense multi-temporal optical imagery widely available. Yet reliable crop classification remains hard throughout Africa and the Global South: spectral similarity between crops, irregular planting and phenological cycles, persistent cloud cover, and rain-fed dryland management all degrade accuracy (Bégué et al., 2020; Mahlayeye et al., 2024). Classical methods such as Random Forest (RF) and XGBoost have proven effective and efficient on engineered features (Mahlayeye et al., 2024; Mann et al., 2026; Mann and Warner, 2017; Mann et al., 2019), but their performance hinges on hand-crafted feature engineering (Maxwell et al., 2018; Belgiu and Drăguț, 2016). Deep learning, recurrent and convolutional temporal models, reduces reliance on hand-crafted predictors and captures phenological structure that summary statistics miss (Zhu et al., 2017; Rußwurm and Körner, 2018; Pelletier et al., 2019), but its generalizability in complex, low-label settings remains uncertain (Pankajakshan et al., 2025).

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region (Belgiu and Drăguț, 2016; Garnot and Landrieu, 2020; Pelletier et al., 2019; Rußwurm and Körner, 2018; Ienco et al., 2017). Such protocols hold out individual pixels or fields but not *space*: training and test observations share the same atmosphere, soils, management calendars, and acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational maps are produced by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint holdout remain rare (Pankajakshan et al., 2025; Jin et al., 2019; Rustowicz et al., 2019). This differs from the bulk of the transfer-learning literature, which adapts a model using target-domain data: labels, unlabeled imagery, or self-supervised pre-training (Ma et al., 2024; Wang et al., 2023; Mai et al., 2025). Our setting is the no-target-access case (domain generalization): models are trained on source tiles and applied to the holdout directly, with no adaptation, so what determines transfer is a property intrinsic to the model rather than a target-specific tuning step.

The Western Cape of South Africa concentrates the operational challenges, smaller fields, rain-fed dryland systems, heterogeneous planting rotations, and persistent winter cloud cover, while offering an unusually rich, high-quality labeled dataset (Radiant Earth Foundation, 2024). This makes it a realistic test bed: rich enough to train and compare modern architectures fairly, yet hard enough to expose the failure modes that matter in practice. We evaluate deep temporal models (CNN-BiLSTM, TabNet, L-TAE, a Transformer encoder, and TempCNN) and deep patch-based 2D/3D CNNs against classical XGBoost,

59 LightGBM, and RF. Critically, we evaluate every model twice: under field-wise
60 cross-validation *within* the training region, and on an adjacent, non-overlapping
61 holdout tile that no training pixel touches. We restrict inputs to multi-temporal
62 Sentinel-2 optical imagery alone, to isolate the spectral-time-series signal and
63 maximize applicability where only optical data are available. All code is public
64 (Mann, 2025a).

65 This study makes five contributions. First, a multi-architecture benchmark
66 (twenty-three classical, deep temporal, and deep patch-based models) showing
67 that in-region versus out-of-region transfer evaluation produce *inverted* rankings
68 on the same data. Second, empirical evidence that models with sparse, axis-
69 aligned feature-selection, the property behind two decades of RF and gradient-
70 boosted-tree dominance, are more stable when moved to a holdout tile, and
71 TabNet inherits it via sparsemax masks. Third, a controlled architectural compar-
72 ison, tested with paired bootstraps, in which adding sparse feature selec-
73 tion improves transfer more than adding attention capacity. We introduce L-
74 TAE-S, a sparse-attention encoder that ties TabNet for the best out-of-region
75 macro-F1, with a field-level variant that matches this accuracy while training
76 in seconds per seed. Fourth, identification of field-level aggregation as a second,
77 substitutable robustness lever that recovers transfer for dense models but is re-
78 dundant for sparse ones. Fifth, a data-efficiency analysis under out-of-region
79 transfer showing the most transferable models are also the most data-efficient,
80 holding near-full holdout accuracy on a quarter of the training fields; as a sup-
81 porting experiment, a controlled comparison of `xr.fresh` time-series features
82 against raw reflectance for XGBoost and TabNet under out-of-region trans-
83 fer. This leads to our central question: across classical machine-learning and
84 deep-learning models, which most reliably classifies crops from multi-temporal
85 Sentinel-2 optical imagery when applied, without target-region labels, to an ad-
86 jacent region outside the training tiles, and what model properties determine
87 whether in-distribution accuracy survives the transfer? Our aim is not to build
88 a single operational model but to test how model families behave under the low-
89 information conditions common across Africa: no target-region labels, a modest
90 single-season training set, and optical imagery only.

91 1.1. Related Work

92 Early crop classification relied on classical learners (RF, SVM, logistic regres-
93 sion, gradient-boosted trees) operating on engineered descriptors (spectral in-
94 dices, temporal summaries) valued for efficiency, interpretability, and robustness
95 on structured data (Maxwell et al., 2018; Belgiu and Drăguț, 2016; Breiman,
96 2001; Chen and Guestrin, 2016; Ke et al., 2017), though their performance
97 depends on feature engineering that can limit transferability across agroeco-
98 logical regions (Wardlow et al., 2007; Mahlayeye et al., 2024). Deep models
99 learn representations directly from reflectance sequences: CNNs capture local
100 spectral-spatial pattern (Kussul et al., 2017), and recurrent and attention ar-
101 chitectures model seasonal phenology (Ienco et al., 2017; Rußwurm and Körner,
102 2018; Garnot and Landrieu, 2020; Rußwurm and Körner, 2020). Other designs
103 combine convolutional feature extraction with temporal modeling (Fan et al.,

2023; Tufail et al., 2025; Bandar and Coşkunçay, 2024), or use deep backbones as feature extractors feeding classical classifiers (Espejo-Garcia et al., 2020; Koklu et al., 2022; Yang et al., 2020). A recurring assumption is that richer learned representations generalize better than compact engineered ones; whether this holds under out-of-region transfer, rather than within-region validation, is less established and is a central question here. Architectures that explicitly target cross-scenario generalization (e.g., Cropformer (Wang et al., 2023)) remain the exception and are rarely benchmarked against classical and deep baselines on a common spatially disjoint holdout. We fill this gap with a systematic comparison under *both* protocols, and additionally evaluate `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting (Mann, 2025b).

2. Data

The study uses a labeled field dataset distributed through Radiant Earth’s MLHub (Radiant Earth Foundation, 2024) and used as the basis of the AI4FoodSecurity (South Africa) challenge (AI4EO, 2021), in an agriculturally productive part of the Western Cape (Fig. 1). Two adjacent Sentinel-2 tiles (`34S_19E_258N`, `34S_19E_259N`) form the training region (4,150 labeled fields over 622 km²; mean field 15.0 ha); a third, immediately adjacent tile (`34S_20E_259N`) is reserved as a holdout for out-of-region transfer evaluation (2,417 fields over 282 km²). The tiles do not overlap, so no pixel or field is shared, and transfer must contend with different soils, geometries, and acquisition conditions. Because the holdout lies within the same agroecological zone and growing calendar, this is a relatively *local* shift, and the gaps we report are best read as a conservative lower bound on degradation under broader transfer. Throughout, “in-region” denotes a field-wise train/validation/test split within the training region; “out-of-region” (OOR) or “holdout” denotes predictions on the adjacent, non-overlapping tile, made without any target-region labels or adaptation. We use “out-of-region” rather than the broader “spatial transfer” because the holdout tests a local shift, not transfer across agroecological zones or years.

We classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing. The classes are heavily imbalanced; lucerne/medics is 43% of training fields and 56% of the holdout (Supplementary Material, Fig. S1(a)), which motivates macro-F1 as the primary metric (Sec. 3.3) and class-imbalance handling during training. Spectrally, wheat and barley overlap strongly, the dominant source of persistent confusion (Supplementary Material, Fig. S2(b)). Their seasonal profiles show the same pattern: lucerne/medics and canola have distinct phenology, whereas the three cereals follow nearly the same curve, and the holdout season runs later than in the training region (Supplementary Material, Sec. S-H, Fig. S8). Sentinel-2 Level-2A surface reflectance was acquired for the January–December 2017 growing season, cloud-masked with `s2cloudless` plus a shadow mask, and composited monthly. Six features per month are used: bands B2, B6, B11, B12, plus EVI and Hue, with EVI and

Study area: Western Cape, South Africa

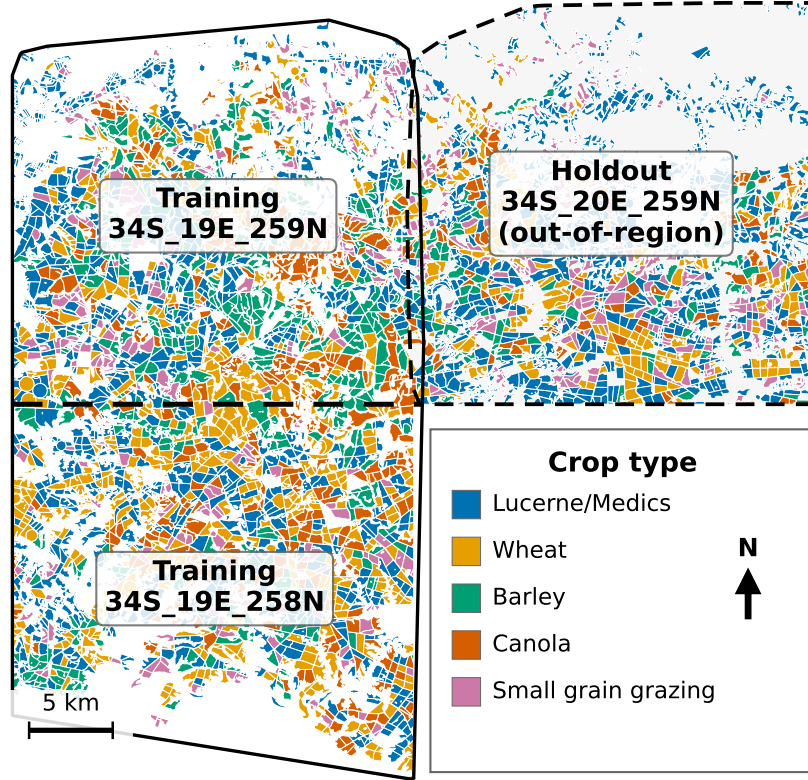


Figure 1: Study area in the Western Cape, South Africa. Two adjacent Sentinel-2 tiles (34S_19E_258N, 34S_19E_259N) form the training region; the immediately adjacent 34S_20E_259N tile (dashed) is held out for out-of-region transfer evaluation. Fields are colored by crop type; the tiles are spatially disjoint, so no pixel or field is shared between training and holdout.

147 Hue computed per scene before compositing, chosen to span the most informa-
 148 tive spectral regions while limiting redundancy among correlated bands (Mann
 149 et al., 2026; Tufail et al., 2025). June 2017 yielded no usable imagery and May
 150 was discarded for cloud, leaving 10 monthly observations, so each pixel is a
 151 60-dimensional time series (6 features $\times$ 10 months). Field polygons were digi-
 152 tized from sub-meter aerial photography and carry a single seasonal crop label
 153 (Radiant Earth Foundation, 2024); all predictions are made at the field level.

154 3. Methodology

155 We compare two model families: classical machine learning on engineered
 156 `xr_fresh` features (9 models) and deep learning (14), the latter comprising deep
 157 temporal models on per-pixel or per-field time series (12) and deep patch-based
 158 models on image patches (2). To prevent leakage, all train/validation/test split-
 159 ting is performed at the field level with a fixed random seed, so no field’s pixels
 160 straddle partitions; pixel- and patch-level predictions are aggregated to a field
 161 label by majority vote (or, for probabilistic models, mean pooling). Computa-
 162 tional cost across paradigms is reported in the Supplementary Material (Sec. S-
 163 C).

164 3.1. Classical Machine Learning

165 We automate feature extraction with the `xr_fresh` toolkit, computing a set
 166 of statistical and temporal descriptors (e.g., autocorrelation, linear trend, day-
 167 of-year of the seasonal maximum, quantiles, complexity) from each pixel’s 10-
 168 month profile, yielding a 174-dimensional per-pixel vector (Mann et al., 2026;
 169 Mann, 2025b); the full feature list is in the Supplementary Material (Sec. S-
 170 A, Table S1). Four baselines: logistic regression, RF (300 trees), LightGBM,
 171 and XGBoost (James et al., 2013; Breiman, 2001; Ke et al., 2017; Chen and
 172 Guestrin, 2016; Friedman, 2001) are trained on this representation, all of them
 173 tree- or margin-based with built-in feature selection or shrinkage; standard
 174 formulations are reproduced in the Supplementary Material (Sec. S-A). We
 175 also report soft-voting and stacked ensembles (RF/XGBoost/LightGBM, with
 176 logistic-regression meta-learner), an Optuna-tuned field-level XGBoost, and a
 177 SMOTETomek-resampled variant; the *Stacking* and *SMOTE Stacked* pair dif-
 178 fers only in resampling, isolating the effect of class rebalancing. At the field
 179 level, pixel features are mean-pooled within each polygon before classification
 180 (Peña-Barragán et al., 2011).

181 3.2. Deep Learning Architectures

182 At the pixel level each observation is a 60-feature temporal sequence. Al-
 183 though the pixel models train on millions of pixels, pixels within a field are highly
 184 autocorrelated and share one label, so the number of statistically independent
 185 units is closer to the number of fields; field-wise splitting keeps evaluation hon-
 186 est at the field scale. We evaluate, as five-seed ensembles, a CNN-BiLSTM
 187 trained with weighted focal loss; TabNet (Arik and Pfister, 2021; Muruganan-
 188 tham et al., 2024), which treats the observations as tabular input and, at each
 189 decision step, uses an attentive transformer to produce a *sparse* feature mask
 190 via sparsemax, selecting a handful of features and ignoring the rest; the L-TAE
 191 (Garnot and Landrieu, 2020), which applies a single learned query over the
 192 temporal sequence; a standard multi-head Transformer encoder (Vaswani et al.,
 193 2017; Rußwurm and Körner, 2020); and TempCNN (Pelletier et al., 2019). De-
 194 tailed equations for these standard models, and the field-level voting rule, are
 195 in the Supplementary Material (Sec. S-A). The Transformer is included as an
 196 informative dense-attention comparison: it increases attention capacity relative

to L-TAE while retaining no explicit sparse feature-selection mechanism. Deep patch-based models (a multi-channel 2D CNN and a 3D CNN) tile each field into 100 m patches resampled to 128×128 and are likewise trained as five-seed ensembles; details in the Supplementary Material (Sec. S-A).

3.2.1. Sparse-Attention L-TAE (L-TAE-S)

To test whether a tree-like, sparse inductive bias can be grafted onto a temporal-attention encoder, we introduce L-TAE-S, which inserts a TabNet-inspired channel-selection gate before the attention heads (Fig. 2). For each of the $n_{\text{head}} = 16$ heads, a per-head linear projection of the embedded, positionally-encoded sequence ($d_{\text{model}} = 128$, key dimension $d_k = 8$) is passed through *sparsemax* (Martins and Astudillo, 2016) to produce a sparse, instance-dependent mask over embedding channels. Formally, for head h and time step $\mathbf{z}_t \in \mathbb{R}^E$,

$$\mathbf{m}_{h,t} = \text{sparsemax}(\mathbf{P}_{h,t} \odot \text{BN}(W_h \mathbf{z}_t)), \quad \tilde{\mathbf{z}}_{h,t} = \mathbf{m}_{h,t} \odot \mathbf{z}_t,$$

209

$$\mathbf{P}_{h+1,t} = \mathbf{P}_{h,t} \odot \max(\gamma - \mathbf{m}_{h,t}, 0), \quad \mathbf{P}_{1,t} = \mathbf{1};$$

head h then forms its keys and values from $\tilde{\mathbf{z}}_{h,t}$ before applying L-TAE attention. Because *sparsemax* projects onto the probability simplex, most channels of $\mathbf{m}_{h,t}$ are exactly zero, and the relaxation prior $\mathbf{P}_{h,t}$ (coefficient $\gamma = 1.5$) discourages successive heads from re-selecting the same channels. These two mechanisms are the sole source of sparsity. The implementation also adds a penalty $\lambda \sum_{h,t} \|\mathbf{m}_{h,t}\|_1$ ($\lambda = 10^{-3}$), but because each *sparsemax* mask sums to one, this term is constant and has zero gradient, so it does not contribute to sparsity. This single block, the only architectural difference from the L-TAE, grafts a tree-like sparse feature-selection bias onto the attention backbone while keeping its temporal modeling, mirroring TabNet’s *sparsemax* masks (Arik and Pfister, 2021). We evaluate L-TAE-S at both the pixel and field level as a five-seed ensemble under the same weighted-focal objective and field-aggregation protocol as the other temporal networks. Its sensitivity to γ , the number of heads, and an explicit sparsity penalty is reported in the Supplementary Material (Sec. S-G, Table S7).

3.3. Performance Metrics

Because the classes are severely imbalanced, plain accuracy is misleading. We adopt macro-F1 ($F1_m$, the unweighted mean of per-class F1) as the primary metric, since it weights every crop equally and penalizes failure on the minority cereals where models differ most. We report three further metrics: Cohen’s Kappa; weighted F1 (wF1, the support-weighted mean of per-class F1); and cross-entropy (Xent), the field-level scoring metric of the AI4FoodSecurity challenge (AI4EO, 2021) from which the dataset is drawn (lower is better, computed on hard label predictions). Formal definitions are in the Supplementary Material (Sec. S-B).

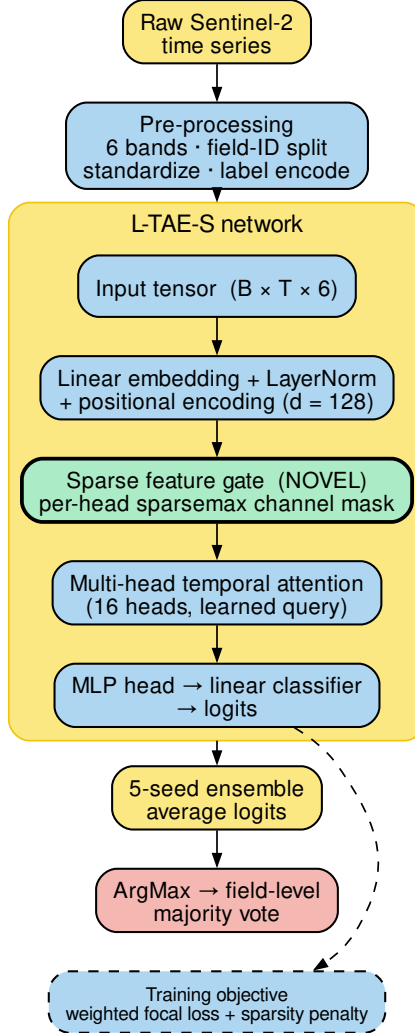


Figure 2: Architecture of the sparse-attention L-TAE-S. Before the multi-head temporal attention, a TabNet-style sparse feature gate (highlighted) applies a per-head **sparsemax** mask that selects a small subset of embedding channels, with a relaxation prior ($\gamma = 1.5$) discouraging heads from re-selecting the same channels. This block is the only architectural difference from the L-TAE; five seeds are ensembled and aggregated to field level.

3.4. Quantifying Uncertainty

We quantify two sources of variability. For sampling uncertainty in the finite holdout, we apply a nonparametric bootstrap over the 2,417 holdout fields: for each model we resample its per-field predictions with replacement $B = 10,000$ times and recompute macro-F1, reporting the 2.5th–97.5th percentile interval as a 95% confidence interval (CI). For between-model differences, we use a paired bootstrap: each replicate resamples the holdout fields once and scores both models on the same fields, giving 95% CIs for Δ macro-F1, $\Delta\kappa$, and Δ accuracy; an exact McNemar test on per-field correctness serves as a check. For optimization noise, the per-seed standard deviation of OOR macro-F1 across the five ensemble members is 0.015–0.037 for the headline temporal models, which is why we report five-seed ensembles throughout. Both sources are small relative to the ranking effects below: every field-bootstrap CI spans only $\approx \pm 0.02$ –0.03 macro-F1, so single-model intervals separate only for differences of ≈ 0.05 or more. Specific comparisons are therefore tested with the paired bootstrap, which is more sensitive because field difficulty cancels.

4. Results

We evaluate every model under two protocols: 1) in-region field-wise validation and 2) out-of-region transfer (OOR) to the adjacent, non-overlapping holdout tile. All reported metrics are field-level. Pixel- and patch-level models are aggregated to field labels, whereas field-level models are trained and evaluated on field-aggregated inputs

4.1. In-Region Validation Overstates Performance and Misranks Models

Table 1 reports both protocols for all models, ranked by OOR macro-F1, with the gap $\Delta = F1_m^{\text{OOR}} - F1_m^{\text{in-region}}$. Two facts stand out. First, *every* model loses accuracy under transfer, but unevenly: from -0.04 (field-level L-TAE-S) to -0.24 (L-TAE at the pixel level). Second, this unevenness reorders the models. In-region the dense temporal pixel nets cluster at the top (L-TAE 0.78, Transformer 0.78, L-TAE-S 0.77, TempCNN 0.77, CNN-BiLSTM 0.72); a practitioner choosing by in-region F1 would pick from this cluster. On the holdout the cluster fractures: the Transformer drops to 0.58 and TempCNN to 0.57 ($\Delta = -0.20$), CNN-BiLSTM to 0.51 ($\Delta = -0.21$), and the in-region leader L-TAE ($\Delta = -0.24$) to 0.54, below every pixel-level tree learner.

The winners under transfer are a different group. The best single model is the TabNet pixel ensemble ($F1_m = 0.60$, $\kappa = 0.54$), which also posts the lowest holdout cross-entropy (4.42) of any model, tied on macro-F1 by both sparse-attention L-TAE-S variants (0.60). The models with the smallest gaps are the sparse and linear learners: field-level L-TAE-S ($\Delta = -0.04$), logistic regression (-0.05), and the pixel-level tree learners. A small gap alone is not evidence of robustness, because it can also come from lower in-region accuracy: field-level L-TAE-S has the smallest gap largely because aggregation lowers its in-region macro-F1 from 0.77 to 0.64 while its holdout score stays at 0.60. We

therefore read gaps together with holdout accuracy. On that joint view, L-TAE-S, a deep temporal attention network whose only change is a learned gate restricting each head to a small channel subset, ties TabNet for the best holdout accuracy at both the pixel and the field level, which supports our thesis. The paired bootstrap confirms which differences are real: the best transferers beat the dense pixel nets by wide, significant margins (TabNet over CNN-BiLSTM, $\Delta = 0.09$, 95% CI [0.07, 0.11]; over L-TAE, $\Delta = 0.06$, [0.04, 0.09]), so the inversion is not holdout noise; at the very top the leaders (TabNet and both L-TAE-S variants) are statistically indistinguishable (all 0.60; TabNet vs. L-TAE-S, $p = 0.73$). We therefore do not claim a single best model; we claim that a sparse-feature-selection group occupies the top of the OOR ranking while the dense temporal nets prized by in-region validation fall significantly below it.

Table 1: In-region versus out-of-region transfer performance of all twenty-three models.

Model	Level	Features	In-reg		Out-of-region (OOR)				
			F1 _m	Δ	F1 _m	95% CI	κ	wF1	Xent
TabNet (raw)	pixel	raw	0.73	-0.13	0.60	.58-.63	0.54	0.70	4.42
L-TAE-S	field	raw	0.64	- 0.04	0.60	.58-.62	0.52	0.69	4.77
L-TAE-S	pixel	raw	0.77	-0.17	0.60	.57-.62	0.53	0.70	4.63
L-TAE	field	raw	0.67	-0.07	0.59	.57-.62	0.47	0.67	5.51
Transformer	pixel	raw	0.78	-0.20	0.58	.55-.60	0.50	0.69	4.94
TempCNN	pixel	raw	0.77	-0.20	0.57	.55-.60	0.49	0.68	4.96
XGBoost	field	xr_fresh	0.70	-0.13	0.56	.54-.59	0.47	0.68	4.89
LightGBM	pixel	xr_fresh	0.62	-0.06	0.56	.54-.59	0.47	0.66	4.85
Logistic Regression	pixel	xr_fresh	0.61	-0.05	0.56	.54-.59	0.47	0.67	4.93
LightGBM	field	xr_fresh	0.68	-0.12	0.56	.53-.58	0.45	0.66	4.99
Voting	pixel	xr_fresh	0.69	-0.13	0.56	.53-.58	0.48	0.66	4.85
Random Forest	pixel	xr_fresh	0.65	-0.10	0.55	.52-.57	0.45	0.65	5.07
XGBoost	pixel	xr_fresh	0.64	-0.09	0.54	.52-.57	0.47	0.66	5.01
L-TAE	pixel	raw	0.78	-0.24	0.54	.51-.56	0.40	0.62	6.42
TempCNN	field	raw	0.67	-0.14	0.54	.51-.56	0.43	0.64	6.07
Stacking	field	xr_fresh	0.64	-0.11	0.53	.51-.56	0.48	0.67	5.32
CNN-BiLSTM	pixel	raw	0.72	-0.21	0.51	.49-.54	0.39	0.61	6.69
CNN-BiLSTM	field	raw	0.64	-0.14	0.51	.48-.53	0.33	0.54	7.86
3D CNN	patch	raw	0.70	-0.20	0.50	.48-.53	0.29	0.54	7.98
SMOTE Stacked	field	xr_fresh	0.69	-0.21	0.49	.46-.51	0.43	0.63	5.37
Multi-Channel 2D CNN	patch	raw	0.63	-0.18	0.45	.42-.47	0.30	0.54	7.62
TabNet (raw, field-avg)	field	raw	0.62	-0.19	0.43	.41-.46	0.31	0.56	7.35
TabNet (xr_fresh)	field	xr_fresh	0.51	-0.15	0.37	.34-.40	0.25	0.53	6.13

Ranked by OOR macro-F1. In-region uses field-wise validation; OOR uses the disjoint holdout tile (34S_20E_259N). Δ is the gap; the 95% CI is the percentile bootstrap on OOR macro-F1 from 10,000 resamples of the 2,417 holdout fields. Xent is the AI4FoodSecurity field-level scoring metric (lower is better). *Features* marks the automated **xr_fresh** statistics (174 features) versus raw reflectance. Bold marks the best value per column.

4.2. Pixel Versus Field Level: Aggregation as Variance Reduction

Several architectures were run at both the pixel and the field level (averaging per-pixel sequences within a field before classification). Table 2 shows that field-level aggregation narrows the gap of every *dense* temporal network (L-TAE $-0.24 \rightarrow -0.07$, TempCNN $-0.20 \rightarrow -0.14$, CNN-BiLSTM $-0.21 \rightarrow -0.14$),

Table 2: Pixel- versus field-level out-of-region transfer.

Architecture	Pixel		Field		Field input
	$F1_m$	Δ	$F1_m$	Δ	
L-TAE-S	0.60	-0.17	0.60	-0.04	raw-avg
L-TAE	0.54	-0.24	0.59	-0.07	raw-avg
XGBoost	0.54	-0.09	0.56	-0.13	xr_fresh
LightGBM	0.56	-0.06	0.56	-0.12	xr_fresh
TempCNN	0.57	-0.20	0.54	-0.14	raw-avg
CNN-BiLSTM	0.51	-0.21	0.51	-0.14	raw-avg
TabNet	0.60	-0.13	0.43	-0.19	raw-avg

Architectures run at both granularities, ranked by field-level OOR $F1_m$. The *Field input* column gives the field-level input (**xr_fresh** statistics for the trees, field-averaged raw sequence for the deep models). Field aggregation narrows the gap for the dense temporal nets but raises their OOR $F1_m$ only for L-TAE, is roughly neutral for the tree learners, and degrades TabNet. Bold marks the best value per column.

but it raises holdout accuracy only for L-TAE ($F1_m$ 0.54 $\rightarrow$ 0.59; paired Δ = 0.05, 95% CI [0.03, 0.08]). For TempCNN (0.57 $\rightarrow$ 0.54) and CNN-BiLSTM (0.51 $\rightarrow$ 0.51) the gap closes mainly because in-region accuracy falls, so aggregation acts as a variance-reduction regularizer that trades in-region fit for stability rather than a reliable accuracy gain. The sparse-attention L-TAE-S follows the L-TAE pattern despite its channel gating: its gap shrinks from -0.17 to -0.04 while OOR $F1_m$ holds at 0.60, so the response to aggregation appears to depend on both the temporal backbone and the input granularity; the sparse gate does not eliminate the benefit of field-level aggregation. TabNet is the informative exception: aggregating its raw inputs to the field level *hurts* ($F1_m$ 0.60 $\rightarrow$ 0.43), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel observation volume it exploits. The tree learners transfer well at either level, and their pixel $\rightarrow$ field change is essentially neutral under transfer (XGBoost 0.54 $\rightarrow$ 0.56). In short, aggregation stabilizes dense models but raises their transfer accuracy only for L-TAE; sparse, tree-like models do not need it.

4.3. Per-Class Performance and Persistent Confusions

Figure 3 breaks accuracy down crop by crop on the holdout for four representative models: the best transferer (TabNet), our sparse-attention L-TAE-S, its dense counterpart L-TAE, and CNN-BiLSTM (the weakest pixel-level temporal network on the holdout). The errors are highly uneven. TabNet and L-TAE-S classify the dominant lucerne/medics and the distinctive canola well ($F1$ 0.88 and 0.81; 0.87 and 0.76), L-TAE less so (0.77 and 0.67), but every model struggles with the cereals and small-grain grazing: even TabNet reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small-grain grazing. Adding the sparsity gate (L-TAE $\rightarrow$ L-TAE-S) on the same backbone improves every class, most on the easier ones (lucerne/medics 0.77 $\rightarrow$ 0.87, canola 0.67 $\rightarrow$ 0.76) and modestly on the cereals (barley 0.47 $\rightarrow$ 0.50, wheat 0.43 $\rightarrow$ 0.47, small-

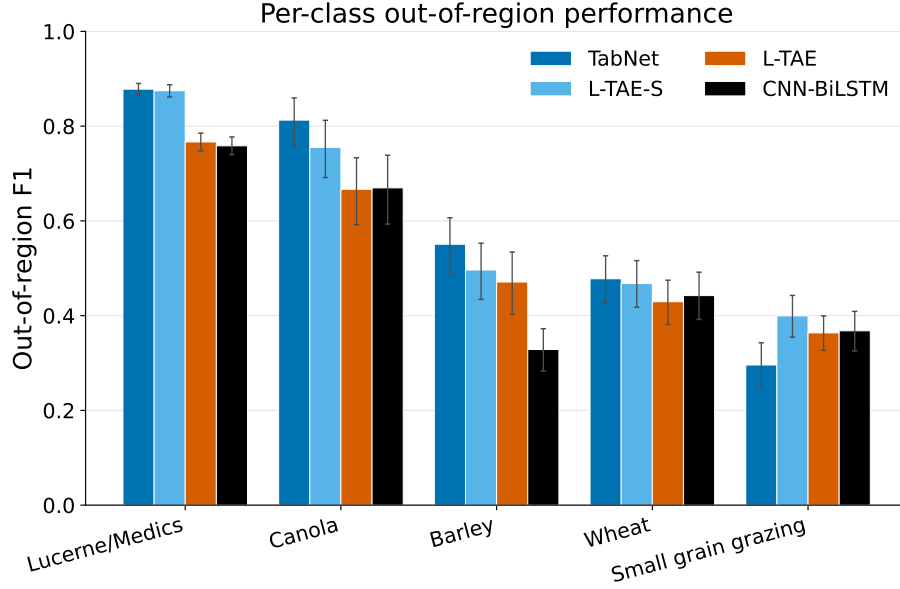


Figure 3: Per-class F1 on the holdout region for four representative models (TabNet, L-TAE-S, L-TAE, and CNN-BiLSTM). TabNet and L-TAE-S classify lucerne/medics and canola best; the cereals (barley, wheat) and small-grain grazing concentrate the errors. The dense L-TAE and CNN-BiLSTM also trail on the easier classes. Error bars are 95% bootstrap CIs over holdout fields.

327 grain grazing 0.36 $\rightarrow$ 0.40). CNN-BiLSTM tracks L-TAE on most classes but is
 328 weakest on barley (0.33), so the dense networks lose accuracy on easy and hard
 329 classes alike, not only on minority classes. The wheat–barley confusion is con-
 330 sistent with their spectral overlap and appears across every model. It appears
 331 that errors are also spatially clustered and do not degrade with distance. Close
 332 to the training tiles, these four models and field-level XGBoost each classify
 333 about 70% of fields correctly; however, in the south-central zone (about 8 to
 334 16 km away) L-TAE and CNN-BiLSTM fall to 36% and 30% while TabNet and
 335 L-TAE-S stay near 70% (Supplementary Material, Sec. S-E, Fig. S7).

336 4.4. Data Efficiency Under Out-of-Region Transfer

337 The most transferable models are also the most data-efficient. We retrain a
 338 representative set on random 75%, 50%, and 25% subsets of the training fields
 339 and re-evaluate on the holdout; because a single subsample can be unrepresenta-
 340 tive, we repeat each fraction over five independent draws and plot the mean
 341 (Fig. 4). Averaged this way, the deep temporal models are essentially flat from
 342 full data down to a quarter of the fields (L-TAE and L-TAE-S vary by under
 343 0.02 macro-F1 across fractions), while the tree, linear, and TabNet baselines lose
 344 only ~ 0.03 – 0.05 , modest given a four-fold cut in training data. The per-fraction
 345 wiggles seen in any single split are within draw-to-draw sampling variance: L-

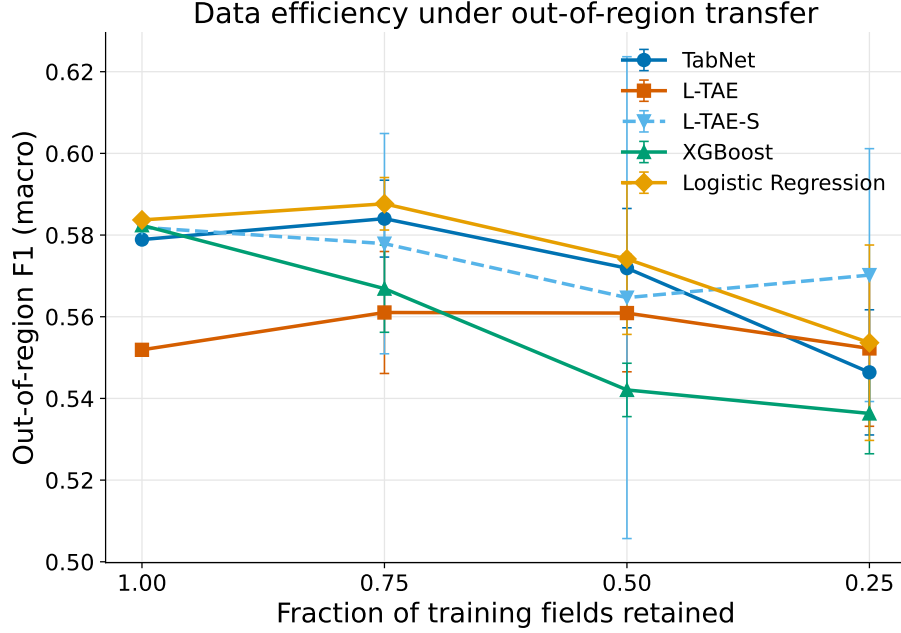


Figure 4: Data efficiency under out-of-region transfer: macro-F1 versus fraction of training fields retained, single training seed, each point the mean over five independent subsample draws (error bars ± 1 s.d.). The deep temporal models (L-TAE, L-TAE-S) are nearly flat to a quarter of the fields; the tree, linear, and TabNet baselines decline modestly. Per-fraction variation is within draw-to-draw sampling noise, largest for L-TAE-S (± 0.06 at 50%), so single-split “dips” are not real. Absolute values run ~ 0.02 – 0.05 below the five-seed levels of Table 1 because a single training seed is used.

346 TAE-S, whose sparse channel gate makes it the most sensitive to *which* fields
 347 are sampled, swings ± 0.06 macro-F1 at the 50% fraction (versus ≤ 0.02 for the
 348 robust learners), so the apparent 50% “dip” of any one draw is sampling noise
 349 rather than a data-efficiency effect. Holdout accuracy is therefore limited far
 350 more by out-of-region transfer than by training-set size, reinforcing that what
 351 governs holdout accuracy is the model’s inductive bias rather than the volume
 352 of training data. Because each subsample is drawn across the full training ex-
 353 tent, this experiment varies the labeled training area (from ≈ 620 to ≈ 155 km²
 354 of fields) rather than the spatial extent of the training region.

355 4.5. Raw Reflectance versus *xr_fresh*

356 The best-transferring classical learners in Table 1 are exactly those fed
 357 *xr_fresh* statistics, so representation and model family are confounded there.
 358 To separate them, we trained a single XGBoost learner once on raw monthly
 359 reflectance and once on the per-pixel *xr_fresh* statistics, holding the FID-wise
 360 splits, preprocessing, and field-level mean pooling fixed (Table 3). The two rep-
 361 resentations are essentially indistinguishable under transfer ($F1_m = 0.59$ raw

Table 3: Raw reflectance versus **xr_fresh** for a single fixed XGBoost learner.

Representation	Split	$F1_m$	wF1	κ
Raw reflectance	In-region	0.73	0.80	0.71
xr_fresh	In-region	0.72	0.76	0.68
Raw reflectance	Out-of-region	0.59	0.69	0.50
xr_fresh	Out-of-region	0.56	0.69	0.50

Identical FID-wise splits, preprocessing, and field-level mean pooling; only the input representation differs. Metrics are field-level macro-F1 ($F1_m$), weighted F1 (wF1), and Cohen’s κ .

vs. 0.56 **xr_fresh**, a difference within the holdout bootstrap interval of Table 1; equal weighted-F1 and κ), and raw reflectance is, if anything, marginally ahead in-region as well. Field-level TabNet, run with the same configuration and seeds on both inputs, points the same way: field-averaged raw sequences transfer better than **xr_fresh** (OOR $F1_m = 0.43$ vs. 0.37, non-overlapping bootstrap intervals; Table 1). For these two learners, **xr_fresh** yields no transfer-accuracy gain over raw reflectance. Its value is operational: a compact, fast-to-fit, agronomically interpretable vector that trains in seconds (Supplementary Material, Sec. S-C). The comparison does not extend to the deep temporal models, because **xr_fresh** reduces each time series to summary statistics with no time axis for L-TAE, TempCNN, or CNN-BiLSTM to attend or convolve over. This is, to our knowledge, the first controlled comparison of **xr_fresh** against raw reflectance under out-of-region transfer.

5. Discussion

The conclusion about which model is “best” depends almost entirely on whether evaluation is in-region or under out-of-region transfer, and a model’s transfer behavior is predictable from its inductive bias.

5.1. Why Tree-Like Inductive Bias Transfers

The clearest pattern is that transfer is strongly associated with inductive bias, and not simply with model family or attention capacity. Figure 5 groups every model by inductive bias and plots its OOR gap: linear models lose the least (mean $\Delta = -0.05$), tree-based models little (mean -0.11), the sparse-attention L-TAE-S sits essentially with the trees (mean -0.11 across its variants), while dense temporal/patch nets lose the most (mean -0.20) and SMOTE-augmented stacking is similarly fragile (-0.21). RF and gradient-boosted trees have been the remote-sensing workhorse for two decades not because they top in-region accuracy (dense nets often do) but because they survive when a model is applied to an unseen region (the operational case) (Belgiu and Drăguț, 2016; Maxwell et al., 2018). A tree decides one feature at a time, choosing the few features that matter for each split and ignoring the rest; this sparsity may reduce reliance on region-specific quirks of phenology, soil, and atmosphere that pull richer models off course. Independent corroboration spans settings: a 3D-CNN collapses under

cross-sensor hyperspectral transfer while a sparse SVM holds (Pankajakshan et al., 2025); RF matches or exceeds a dense CNN-LSTM on label-scarce Ghana fields (Rustowicz et al., 2019); a maize model drops 16 points moving across East Africa (Jin et al., 2019); and bagged/boosted trees out-transfer an elaborate instance-reweighting scheme from the U.S. to northwest China (Mai et al., 2025). TabNet, the best transferer here, is a neural network built to behave like a tree: a small learned gate picks a handful of features at each step (Arik and Pfister, 2021), so it clusters with the trees and inherits their transferability. By contrast, CNN-BiLSTM, TempCNN, L-TAE, and the Transformer consume the whole temporal sequence at once with no feature selection. At the pixel level this leaves them fitting the training region tightly: they post the largest in-region-to-holdout gaps in the benchmark, with L-TAE dropping furthest ($\Delta = -0.24$). This is consistent with evidence that the dates such models attend to are themselves contingent on the training region (Obadic et al., 2022). Dense attention is not intrinsically poor under out-of-region transfer: the pixel-level L-TAE loses 0.24 macro-F1, but the field-level L-TAE reduces this gap to 0.07 and raises holdout macro-F1 to 0.59 (Sec. 4.2), so it is the pixel-level dense models specifically that transfer worst.

The Transformer and L-TAE-S together separate attention capacity from feature selection on the same pixel-level task. The Transformer deploys far richer attention than L-TAE’s single query and does transfer better ($F1_m$ 0.58 vs. 0.54; paired $\Delta = 0.04$, 95% CI $[0.02, 0.06]$), so capacity is not irrelevant. Adding sparse feature selection to the smaller L-TAE helps more: L-TAE-S reaches 0.60 ($\Delta = 0.06$, $[0.04, 0.08]$; $\Delta\kappa = 0.13$) and matches or exceeds the Transformer (Δ macro-F1 0.02, $p = 0.06$; $\Delta\kappa$ and Δ accuracy 0.02, both significant). At the field level the gate’s macro-F1 gain over L-TAE is not significant (0.60 vs. 0.59), but κ and accuracy improve ($\Delta = 0.04$ and 0.05, both significant). L-TAE-S and TabNet, the two models that explicitly select features before computing with them, are the best transferers. Within the architectures tested, sparse feature selection is therefore associated with larger and more consistent transfer gains than added attention capacity; this is consistent with, but does not prove, feature selection as the property that matters most. The L-TAE-S experiment also shows that this is *not* a tree-versus-deep tradeoff: the same selection step used by tree splits can be grafted onto a dense architecture at the cost of a single block, and L-TAE-S ties TabNet for the best holdout accuracy. This result does not hinge on the chosen hyperparameters: across 22 of 23 field-level settings of γ and the number of heads, L-TAE-S scores between 0.58 and 0.62, and at both pixel-level settings tested it still improves significantly on L-TAE, although fewer heads transfer slightly better (Supplementary Material, Sec. S-G). That parity is not accuracy alone: at the pixel level L-TAE-S reaches TabNet’s best-in-class holdout accuracy in roughly 40% of the per-seed training time (24 vs. 61 min), and its field-level variant is both faster and more accurate than any field-aggregated TabNet (Supplementary Material, Sec. S-C). The next generation of operational crop classifiers need not choose between the expressivity of deep learning and the transfer-robustness of trees.

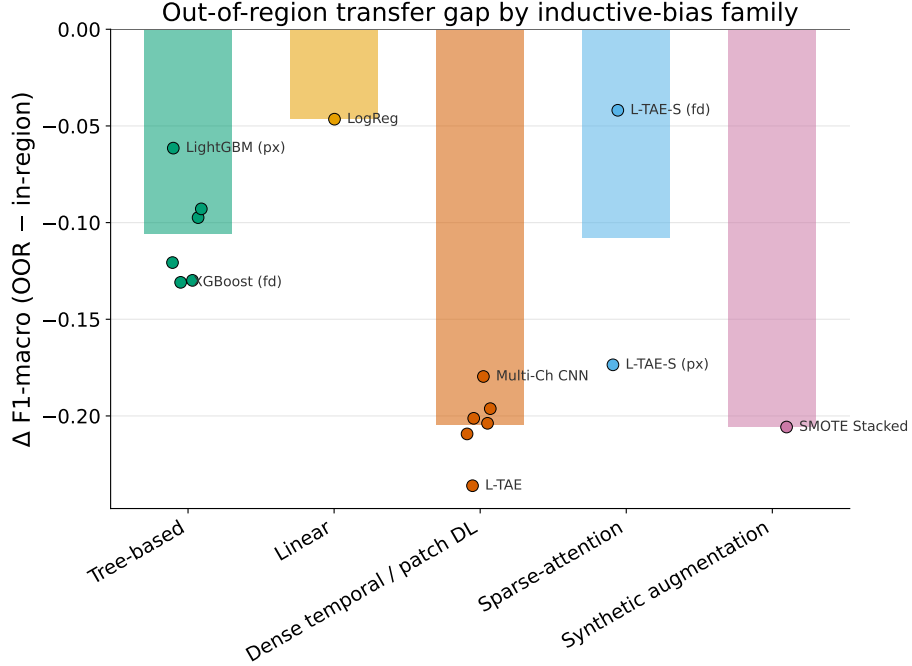


Figure 5: Out-of-region transfer gap (Δ macro-F1) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small $|\Delta|$); dense temporal/patch nets and synthetic augmentation overfit the training region. L-TAE-S, which adds sparse feature selection to a dense attention backbone, tracks the tree-based models. Bars are family means; points are individual models.

5.2. Two Robustness Levers

Two caveats sharpen rather than weaken the story, and together they point to how we can choose the best models. First, it is *pixel-level* TabNet that wins; the field-aggregated TabNet variants are among the worst transferers (Table 1), because aggregation removes the per-pixel observation volume the sparse model exploits. Second, dense temporal attention transfers well *when* applied at the field level: L-TAE’s pixel-level gap shrinks from -0.24 to -0.07 under aggregation, and L-TAE-S ends with an even smaller gap (Table 2); for TempCNN and CNN-BiLSTM aggregation narrows the gap without raising holdout accuracy. These point to two distinct, partly substitutable directions. The first lever is built-in feature selection. The model itself picking a few features at a time (trees, TabNet, and now L-TAE-S); this is intrinsic robustness paid for at design time. The second is variance reduction imposed from outside via field-level aggregation, which averages away region-specific pixel noise before the classifier sees it. Which provides external robustness available even when the architecture is fixed. A model with the first lever gains nothing, and may lose observation volume, from the second; this is exactly why TabNet prefers pixels. These two properties can be combined. L-TAE-S adds a sparse gate to

457 a temporal-attention backbone and then tolerates field aggregation: its holdout
 458 macro-F1 stays at 0.60 while per-seed training time falls from 24 min to 18 s.
 459 Practitioners can read this as a flowchart: if free to choose or modify the model,
 460 build feature selection in (lever 1); if constrained to an off-the-shelf dense ar-
 461 chitecture, aggregate to the field level (lever 2), which stabilizes transfer even
 462 when it does not raise accuracy. Two further controlled comparisons confirm
 463 the ranking is not an artifact of ensembling or imbalance handling; both shift
 464 absolute scores but never reorder the robust and fragile groups, with the full
 465 tables in the Supplementary Material (Sec. S-D). Seed-ensembling gives only
 466 a small, monotone lift that leaves the ordering intact. A single-seed L-TAE-S
 467 ($F1_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field
 468 ensembles (Table S4). And the imbalance-handling objective moves absolute
 469 scores but not the ranking: the robust L-TAE-S is almost invariant across five
 470 loss and sampler settings ($F1_m = 0.59$ – 0.60), whereas the fragile CNN-BiLSTM
 471 swings widely (0.41 – 0.56) and even at its best stays below L-TAE-S’s worst
 472 (Table S5).

473 5.3. Automated Features, Patch Models, and Persistent Difficulty

474 The classical pipelines paired with `xr_fresh` were not only robust but cheap,
 475 transferring as well as the best deep models (OOR $F1_m$ 0.55–0.56) while train-
 476 ing in minutes rather than hours (Supplementary Material, Sec. S-C); the con-
 477 trolled comparisons for XGBoost and TabNet (Sec. 4.5) show that this value
 478 is operational rather than an accuracy edge over raw reflectance. Patch-based
 479 convolutional models did not exceed pixel-level models and transferred poorly
 480 (3D CNN $F1_m = 0.50$; multi-channel 2D CNN 0.45). We tested one configura-
 481 tion of each, so this should not be read as a property of patch CNNs in general.
 482 Several design factors may also contribute to it. At 10 m resolution a 100 m
 483 patch holds only $\sim 10 \times 10$ pixels, and upsampling to 128×128 adds interpo-
 484 lation rather than detail. After upsampling, the convolutional receptive fields
 485 span only one to two native pixels before global pooling or flattening, so little
 486 spatial structure is learned. The two models also fuse time differently: the 2D
 487 CNN stacks all band-month images as input channels, discarding temporal order,
 488 while the 3D CNN convolves across months. Lack of sparse feature selection
 489 may also play a role. Our experiments do not separate these factors; patch-size
 490 and architecture sweeps would be needed to do so. Finally, small-grain grazing,
 491 barley, and wheat are the hardest classes for every model (Fig. 3) and account
 492 for most accuracy lost under transfer; that even robust models struggle here,
 493 and that the dense networks trail on the easier classes as well, indicates the
 494 limitation is partly intrinsic spectral similarity and partly the transfer problem
 495 itself, not class imbalance alone. Much of this confusion is among the cereals
 496 themselves rather than with the other crops. Merging barley, wheat, and small-
 497 grain grazing into a single cereal class, without retraining, raises out-of-region
 498 accuracy from 0.73 to 0.84 for TabNet and from 0.71 to 0.83 for L-TAE-S, with
 499 the merged class reaching an F1 of about 0.77. Under the merged scheme the
 500 sparse models remain at the top of the ranking and the dense pixel networks

at the bottom (Supplementary Material, Sec. S-F, Table S6). The cereals’ seasonal profiles nearly coincide, and the later holdout season shifts them by more than they differ from one another, which adds to their confusion under transfer (Supplementary Material, Sec. S-H).

5.4. Scope and Practical Relevance

The absolute accuracies reported here (out-of-region macro-F1 of at most 0.60) are too low for an operational crop map, and producing one is not our aim. The study is a controlled experiment on how model families behave under the low-information conditions typical of much of Africa: no labels in the target area, a modest labeled training set from a single season, and optical imagery only. Under these constraints the useful result is relative: which models lose the least when moved to unlabeled land, and how to choose among them when no target labels exist to check that choice. Three practical points follow. First, a model’s accuracy within the training region is a poor guide to how it will perform elsewhere, so when no labels exist in the target area, the safer choice is a model with built-in sparse feature selection, such as TabNet or L-TAE-S, or a dense model whose predictions are aggregated to the field level. Second, the crop-specific results are more useful than the overall score. Lucerne/medics, the dominant crop, and canola transfer well: TabNet and L-TAE-S reach out-of-region F1 of 0.87 to 0.88 for lucerne/medics and 0.76 to 0.81 for canola, so these crops can be mapped with reasonable confidence in unlabeled neighboring areas. The cereals do not: barley (0.50 to 0.55), wheat (0.47 to 0.48), and small-grain grazing (0.30 to 0.40) remain confused with one another for every model, consistent with their overlapping spectral and phenological profiles. In this setting cereal maps are better reported as a combined cereal class, or improved with additional data such as SAR, than separated with optical time series alone. The sparse gate helps across all five crops (L-TAE to L-TAE-S), but it does not close the cereal gap. Third, the robust models are cheap to train: the classical pipelines need minutes and field-level L-TAE-S seconds per seed, which suits settings with limited computing resources. Adding SAR, multiple seasons, or a small number of target-region labels is the natural route from this benchmark toward operational accuracy.

5.5. Limitations

The study covers one growing season (2017) and one agroecological zone, with a holdout tile *adjacent* to the training tiles. Holdout fields lie 0 to 24 km from the nearest training field (median 11 km). The models are trained on ≈ 620 km² of labeled fields within a $\approx 25 \times 45$ km region and are used to predict in a $\approx 24 \times 24$ km tile. This is a relatively local shift, so the gaps we report most likely *understate* the degradation under cross-regional or cross-year transfer and are best read as a conservative lower bound. Cross-year and cross-ecozone transfer could not be tested, because the benchmark labels only these three tiles in a single season. Importantly, in South Africa, like much of Africa, training data is sparse if not non-existent. Multi-year, multi-region holdouts

are needed to confirm that the inductive-bias ordering holds under larger shifts. Our conclusions therefore apply to out-of-region transfer between neighboring tiles at these distances and scales. Training-set size is the one scale we vary: the most transferable models keep near-full holdout accuracy on a quarter of the training fields (Sec. 4.4). The labels derive from a public dataset rather than our own campaign, and we use Sentinel-2 optical data only; a deliberate scoping choice that nonetheless limits the achievable ceiling for optically similar cereals, making SAR fusion a clear next step.

6. Conclusion

In-region validation, the protocol the crop-classification literature reports almost universally, misranks twenty-three Sentinel-2 classifiers when they are applied to an adjacent, non-overlapping holdout tile. Dense temporal networks that lead under conventional evaluation fall sharply under transfer, the in-region leader (L-TAE) dropping below every pixel-level tree learner, while parsimonious models such as tree ensembles, logistic regression, and a TabNet ensemble retain rank and accuracy. The common thread is the same property that made tree ensembles the remote-sensing default for two decades: a sparse, axis-aligned feature-selection bias that commits to a few robust signals and ignores the rest. TabNet inherits it through sparsemax masks. A Transformer with far greater attention capacity transfers better than the lightweight L-TAE, but adding a sparse gate to L-TAE helps more, pointing to sparsity rather than capacity as the stronger lever. Grafting that lever onto an attention model (L-TAE-S) ties TabNet for the best transfer at a fraction of its training cost and, at the field level, keeps that accuracy while training in seconds per seed. Field-level aggregation is a second, partly substitutable lever: it narrows the transfer gap of dense temporal networks, raising holdout accuracy for L-TAE and L-TAE-S, but hurts TabNet. We therefore recommend the following: evaluate on an out-of-region holdout before declaring one architecture better than another; prefer models with a built-in feature-selection bias when mapping nearby unlabeled areas; and, if dense temporal models are required, aggregate their predictions to the field level. The conditions that make the Western Cape difficult are the conditions under which agricultural remote sensing must operate across much of the Global South, and evaluating models the way they are actually deployed, applied to regions where no labels were collected, is what will make remote sensing deliver where it matters most.

CRediT authorship contribution statement

Michael L. Mann: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Sairam Venkatachalam:** Software, Formal analysis, Writing – original draft, Investigation. **Disha Kacha:** Software, Formal analysis, Investigation, Writing – original draft. **Devarsh Sheth:** Software, Formal

analysis, Investigation, Writing – original draft. **Ryan Engstrom:** Supervision, Methodology, Writing – review & editing. **Amir Jafari:** Methodology, Methodology, Supervision, Writing – review & editing.

Ethics statement

This article does not contain any studies with human participants or animals performed by any of the authors. The study is based solely on publicly available satellite imagery (Sentinel-2) and field-boundary crop labels, so no ethical approval was required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The processed dataset supporting this study is openly available on IEEE DataPort at <https://dx.doi.org/10.21227/n69k-r458> (DOI: 10.21227/n69k-r458). The source code is available at https://github.com/mmann1123/South_Africa_Crop_Comp (Mann, 2025a). The study builds on the ground-reference crop labels of the Western Cape Department of Agriculture and Radiant Earth Foundation, “Crop Type Classification Dataset for Western Cape, South Africa”, v1.0, Radiant MLHub (DOI: 10.34911/rdnt.j0co8q), used under the CC-BY-4.0 license; Sentinel-2 imagery was extracted via Google Earth Engine.

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